MATD12 – REPRESENTATION THEORY

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1. Lecture 1

1.1. **Setting the scene.** In many practical situations where we invoke groups in mathematics, it is not the group that is of importance but rather how it acts on other objects. Representation theory aims to ask the question: "What are all **things** that a given group can act on?" There are a few possible candidates for **things**. We will focus on vector spaces. Fix a field k.

Given a group G and a vector space V, we say that a homomorphism $\rho : G \to \mathrm{GL}(V)$ is a *representation* of G in V. The *degree* of ρ is defined to be $\dim(V)$ which may or may not be infinite.

An important remark is that we will often fix a basis of V and consider ρ as a homomorphism from G to the matrix group $GL_n(k)$ where n is the dimension of V. We will use these definitions interchangeably and without prior warning.

Whenever one defines an object in mathematics, one is morally obligated to also introduce a notion of morphisms. Given two representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$ we say that a linear map $\varphi : V \to W$ is a *morphism of representations* if for each $g \in G$ the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "\rho_g"] \arrow[d, "\varphi"'] & V \arrow[d, "\varphi"] \\ W \arrow[r, "\rho'_g"'] & W \end{tikzcd}$$

In particular, if φ is bijective, we shall say that ρ and ρ' are *isomorphic* representations.

Example 1.1. We will now give a non-example to show that an isomorphism of representations is stronger than an isomorphism of the underlying vector space. Consider the two representations on $\mathbb{C}$ of $C_2 = \langle x \mid x^2 = 1\rangle$ given by

$$\rho_1 : x \mapsto 1 \qquad \text{and} \qquad \rho_2 : x \mapsto -1.$$

Notice that the underlying vector spaces are the same, but suppose for contradiction that there exists an isomorphism of representation $\varphi : \mathbb{C} \to \mathbb{C}$. There exists a nonzero $\lambda \in \mathbb{C}$ such that $\varphi = \lambda I$. Since φ was assumed to be a G-homomorphism it must be the case that

$$\begin{gathered} \rho_1(x)(\varphi(1)) = \varphi(\rho_2(x)1) \iff \\ \lambda = -\lambda \end{gathered}$$

which is a contradiction. Thus, these two representations are not isomorphic.

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1.2. **Representations as modules.** For the reader that knows what a module is, we see that the definition of a morphism above is indeed a natural one. Notice that given a group G we can form the vector space $k[G]$ for which the basis consists of all elements of G. We can define a multiplication on $k[G]$ by extending the group multiplication linearly to make $k[G]$ into an algebra. Then, we see that we can identify representations with $k[G]$-modules. Given a representation $\rho : G \to \mathrm{GL}(V)$ we give V the structure of a $k[G]$-module by imposing that

$$g \cdot v = \rho_g(v).$$

Similarly, given a $k[G]$-module V, we define a homomorphism

$$\begin{aligned} \rho : G &\longrightarrow \mathrm{GL}(V) \\ g &\longmapsto (v \mapsto g \cdot v). \end{aligned}$$

Under this identification, a morphism of representations is simply a homomorphism of $k[G]$-modules.

This perspective of representations as $k[G]$-modules will be important later on.

1.3. **First examples and further definitions.**

Example 1.2. Let us look at some first few examples.

(a) The map $\rho : \mathbb{R} \to \mathbb{R}^\times$ given by $\rho_t = e^t$ is a 1-dimensional representation of $\mathbb{R}$.
(b) The map $\rho : \mathbb{R} \to \mathrm{GL}(C(\mathbb{R}))$ given by $\rho_t(f)(x) = f(x+t)$ is an infinite-dimensional representation of $\mathbb{R}$.
(c) A representation of $\mathbb{Z}$ is simply an invertible linear operator $T : V \to V$.
(d) A rperesentation of $\mathbb{Z}/n\mathbb{Z}$ is a linear operator $T : V \to V$ such that $T^n = \mathrm{id}$.

Given a representation $\rho : G \to \mathrm{GL}(V)$ and a subspace $W \subseteq V$ for which $\rho_g(W) \subseteq W$ for all $g \in G$ we say that $\rho\,|_W : G \to \mathrm{GL}(W)$ is a subrepresentation of V. From the module point of view a subrepresentation is simply a $k[G]$-submodule.

Exercise 1.3. Prove that $\rho_g(W) \subseteq W$ for all $g \in G$ is equivalent to $\rho_g(W) = W$ for all $g \in G$. If we fix $g \in G$, is it true that $\rho_g(W) \subseteq W$ if and only if $\rho_G(W) = W$? What if W is finite-dimensional?

A nonzero representation $\rho : G \to \mathrm{GL}(V)$ is called *irreducible* if it has no proper nonzero subrepresentations.

Example 1.4. Consider the following examples.

(a) We can now ask whether the representation $\rho : \mathbb{R} \to \mathrm{GL}(C(\mathbb{R}))$ given by $\rho_t(f)(x) = f(x+t)$ is irreducible. The answer is obviously no. For instance, the subspace of 1-periodic functions is a subrepresentation. Let us determine all the 1-dimensional subrepresentations of ρ. If $f(x) \in C(\mathbb{R})$ generates a 1-dimensional subrepresentation, then there exist constants $\lambda(t)$ for each $t \in G$ such that $f(x+t) = \lambda(t) \cdot f(x)$ and so in particular $f(t) = \lambda(t) \cdot f(0)$. If $f(0) = 0$ then $f \equiv 0$ and thus does not generate a 1-dimensional subspace. If, on the other hand, $f(0) \neq 0$, then $\lambda(t) = f(t)/f(0)$ is continuous and satisfies that $\lambda(t + t') = \lambda(t)\lambda(t')$. One of the later exercises implies that $f(t) = f(0) \cdot e^{at}$ for some $a \in \mathbb{R}$. It is easy to check that all functions on this form do indeed generate a 1-dimensional subrepresentation and our characterisation is therefore completed.

(b) Which representations $\rho : \mathbb{Z} \to \mathrm{GL}_n(\mathbb{C})$ are irreducible? If $n = 1$, then ρ is certainly irreducible. We claim that if $n > 1$, then ρ cannot be irreducible. Indeed, we recall from linear algebra that ρ_1 will have an eigenvalue.

(c) Consider the representation $\rho : S_3 \to \mathrm{GL}_2(\mathbb{R})$ which is given by letting S_3 act on the vertices of a regular triangle in the plane. Clearly this action does not preserve any lines through the origin and is hence irreducible.

Consider now the complexification $\rho^{\mathbb{C}} : S_3 \to \mathrm{GL}_2(\mathbb{C})$. A more interesting question is whether this representation is irreducible. Suppose for the sake of contradiction that there exists a S_3-invariant 1-dimensional subspace $\langle v \rangle \subseteq \mathbb{C}^2$. Let $r \in S_3$ correspond to a rotation and $s \in S_3$ correspond to a reflection. Since $\rho_r^3 = \mathrm{id} = \rho_s^2$ we get that both ρ_r and ρ_s are diagonalisable and since $\rho_r \neq \mathrm{id}$ we deduce that there exists a third root of unity ϵ such that $\rho_r(v) = \epsilon v$ and $\rho_s(v) = \pm v$. This gives us that $\rho_{rs}(v) = \pm\epsilon v$, but rs corresponds to a reflection, so $\pm 1 v = \pm \epsilon v$. Thus $v = 0$, a contradiction. Hence $\rho^{\mathbb{C}}$ is irreducible.

Exercise 1.5. Prove that any continuous 1-dimensional representation of $\mathbb{R}$ is given by $f(t) = e^{at}$ for some $a \in \mathbb{R}$.

Exercise 1.6. Let us expand on an example above.

(a) Prove that any linear operator $A : V \to V$ where $\dim_{\mathbb{C}}(V) > 1$ has an eigenvalue.

(b) Show that the representation $\rho : \mathbb{Z} \to \mathrm{GL}_2(\mathbb{R})$ given by

$$\rho_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

is irreducible.

(c) Tweak the proof above, to show that any representation $\rho : \mathbb{Z} \to \mathrm{GL}_3(\mathbb{R})$ is reducible.

(d) We can do better. Show that for any $n > 2$, a representation $\rho : \mathbb{Z} \to \mathrm{GL}_n(\mathbb{R})$ is reducible.

1.4. **Maschke's theorem and indecomposable representations.** Our goal is to eventually be able to understand all representations of a given group (for sufficiently nice groups and over sufficiently nice fields). In a perfect world, we would be able to say that the study of irreducible representations lets us understand arbitrary representations completely. It turns out that we live in an (almost) perfect world.

Example 1.7. For a more serious example, we will consider the real representations of the dihedral group

$$D_8 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle.$$

We first find all the 1-dimensional representations. Suppose that $\phi : D_8 \to \mathbb{R}^\times$ is a representation. From the relations above one deduces that $\phi(r)^2 = \phi(s)^2 = 1$. Conversely each choice of $\phi(r)$ and $\phi(s)$ as ± 1 gives a representation of degree 1.

We also notice that there is degree 2 representation coming from the action of D_8 on the square situated in the plane. This representation $\rho : D_8 \to \mathrm{GL}(\mathbb{R}^2)$ can be given by

$$\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Can we find any more representations of D_8? The answer is yes for a silly reason. Think of the action of D_8 on a square situated in the xy-plane of $\mathbb{R}^3$. Then this new representation can be given by

$$\rho'(r) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \rho'(s) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Repeating this trick, we can construct arbitrarily large representations.

In the last example, we saw that we can "glue together" representations. This motivates the following definition. Given two representations $\rho_1 : G \to \mathrm{GL}(V)$ and $\rho_2 : G \to \mathrm{GL}(W)$ we can form their *direct sum* $\rho_1 \oplus \rho_2 : G \to \mathrm{GL}(\mathrm{V} \oplus \mathrm{W})$ defined by

$$\rho(s)(v, w) = (\rho_1(s)v, \rho_2(s)w).$$

If ρ is a representation of G such that there do not exist proper subrepresentations ρ_1 and ρ_2 with $\rho_1 \oplus \rho_2 \cong \rho$ we say that ρ is *indecomposable*.

Exercise 1.8. Show that any irreducible representation is indecomposable.

Example 1.9. Let us consider two cases where the notion of irreducible and indecomposable representations differ.

(a) Consider the representation of $C_2 = \langle x \mid x^2 = 1 \rangle$ over the field $\mathbb{F}_2$ given by $\rho : C_2 \to \mathrm{GL}(\mathbb{F}_2^2)$ and

$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

This is a representation since

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is **not irreducible** since the subspace generated by $(1, 0)$ gives a subrepresentation. However, one can show that this is the only subrepresentation of ρ and thus ρ is **indecomposable**.

(b) Consider the related representation $\rho : \mathbb{Z} \to \mathrm{GL}_2(\mathbb{C})$ given by

$$\rho_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Similarly to above, one can show that ρ_1 is not irreducible, yet indecomposable.

This is troubling, since understanding all representations is equivalent to understanding the indecomposable representations. On the other hand, irreducible representations will be easier for us to deal with. These two notions converge when G is finite and $\mathrm{char}(k) = 0$. Please note that we only use that $|G| \neq 0$ in k in our proof and hence it suffices to assume that $\mathrm{char}(k) \nmid |G|$.

Theorem 1.10 (Maschke)**.** *Let k be a field of characteristic 0 and G a finite group. If $\rho : G \to \mathrm{GL}(V)$ is indecomposable, then it is irreducible.*

We will first give a proof of a geometric flavour that only works when $k = \mathbb{R}$ or $\mathbb{C}$ and V is finite-dimensional.

Proof. We will prove the contrapositive. Suppose that $W \subseteq V$ is a proper nonzero subrepresentation. Since V is finite-dimensional over $\mathbb{R}$ or $\mathbb{C}$ it admits an inner product B. Using this inner-product we could define $W^\perp$ and then clearly $V = W \oplus W^\perp$ but the problem is that $W^\perp$ need not be a subrepresentation (i.e. G-invariant). To fix this, we will use an averaging trick. Consider the inner product $B' = \sum_{g\in G} B(\rho_g v, \rho_g w)$ which is well-defined since G is finite. Clearly $B'(\rho_g v, \rho_g w) = B'(v, w)$ for all $g \in G$ (the terms in the sum are simply permuted). Now, let $W^\perp$ be the orthogonal complement of W with respect to this inner product B'. Then, it follows that $W^\perp$ is G-invariant since for any $v \in W^\perp$ and any $w \in W$:

$$B'(\rho_g v, w) = B'(v, \rho_{g^{-1}} w) = 0$$

since W is G-invariant. Thus, $V = W \oplus W^\perp$ is a reducible representation. The theorem follows. □

Exercise 1.11. Deduce the following for any finite group G using our geometric proof of Maschke's theorem:

(a) If $\rho : G \to \mathrm{GL}_n(\mathbb{R})$ is a representation, then there exists a basis of $\mathbb{R}^n$ such that $\mathrm{Im}\rho \subseteq O_n$.
(b) If $\rho : G \to \mathrm{GL}_n(\mathbb{C})$ is a representation, then there exists a basis of $\mathbb{C}^n$ such that $\mathrm{Im}\rho \subseteq U_n$.

Exercise 1.12. Explain why any finite-dimensional representation of a finite group over a field of characteristic 0 can be decomposed as a direct sum of irreducible representations. Does the same result hold for infinite-dimensional representations? For infinite groups? For fields of positive characteristic?

2. Lecture 2

2.1. **A fun exercise.** Recall the complex representation

$$\rho : \mathbb{R} \longrightarrow \mathrm{GL}(C(R))$$
$$\rho_t(f)(x) = f(x+t).$$

Last time, we showed that all 1-dimensional subrepresentations are given by $\langle e^{ax}\rangle$. For 2-dimensional subrepresentations, we can for instance find $\langle e^{ax}, xe^{ax}\rangle$.

Exercise 2.1. Is it true that the only n-dimensional indecomposable subrepresentation is

$$\langle e^{ax}, xe^{ax}, x^2e^{ax}, \ldots, x^{n-1}e^{x}\rangle.$$

2.2. **A second proof of Maschke's theorem.** We will now give a proof for Mascke's theorem (Theorem 1.10) that works in general.

Proof. Let G be a finite group, k be a field of characteristic 0 and $\rho : G \to \mathrm{GL}(V)$ a representation. Suppose that $W \subseteq V$ is a subrepresentation.

From linear algebra, we know that there exists a projection $\pi' : V \to V$ such that $\pi'(V) \subseteq W$ and $\pi'|_W = \mathrm{id}_W$. We will first try to find a projection which is a morphism of representations. This can be done by an averaging trick. Consider the linear map $\pi : V \to V$ given by

$$\pi(v) = \frac{1}{|G|} \sum_{g\in G} \rho_g \circ \pi'(\rho_{g^{-1}} v).$$

This is clearly a projection and a morphism of representations from $V \to V$. You will show in the exercises that the category of representations is abelian so $W' = \ker(\pi)$ is a subrepresentation of V and since π is a projection $W \oplus W' \cong V$ which proves that V is decomposable. This finishes the proof. $\square$

Exercise 2.2. In this exercise, we will verify the computations in the above proof that were left out.

(a) Prove that if $W \subseteq V$ are vector spaces then there exists a projection $\pi' : V \to V$ such that $\pi'(V) \subseteq W$ and $\pi'|_W = \mathrm{id}_W$.
(b) Prove that π above is a projection.
(c) Prove that π is a morphism of representations.

Exercise 2.3. It turns out that the category of representations of a group G over a field k is abelian. You are not expected to know what this means. Instead, prove that if φ is a morphism of representations, then $\ker(\varphi)$ and $\mathrm{im}(\varphi)$ are representations.

2.3. **Schur's lemma.** In this section, our treatment will deviate somewhat from the lecture. The main theorem of this sections is the following.

Theorem 2.4 (Schur)**.** *Let (V, ρ) and (V', ρ') be representations of a group G over a field k and let $\varphi : V \to V'$ be a morphism of representations. Then:*

(i) if V is irreducible, then φ is injective.
(ii) if V' is irreducible, then φ is subjective.
(iii) if $(V, \rho) = (V', \rho')$ is irreducible and finite dimensional and k is algebraically closed, then $\varphi = \lambda I$ for some $\lambda \in k^\times$.

Proof. Part (i) follows from the fact that $\ker(\varphi)$ is a subrepresentation of V and part (ii) follows from the fact that $\mathrm{im}(\varphi)$ is a subrepresentation of V' using Exercise 2.3.

For part (iii), assume that $\varphi : V \to V$ is a nonzero endomorphism of an irreducible representation over an algebraically closed field k. Since k is algebraically closed, there exists an eigenvalue $\lambda \in k$ of φ. In particular, $\varphi - \lambda \cdot \mathrm{id} : V \to V$ is a morphism of V which is not injective. Hence, by part (i), $\varphi - \lambda \cdot \mathrm{id} = 0$ which finishes the proof. $\square$

What is the upshot of Schur's lemma? Let V and W be representations of a group G over a fixed base field k. We will let $\mathrm{Hom}_G(V, W)$ denote the vector space of G-homomorphisms from V to W and $\mathrm{End}_G(V)$ the $k[G]$-algebra of G-endomorphisms of V. Then:

- If V and W are nonisomorphic representations, then $\dim \mathrm{Hom}_G(V, W) = 0$.
- If $V \cong W$, then $\dim(\mathrm{Hom}_G(V, W)) > 0$.
- If V is irreducible, then the k-algebra $\mathrm{End}_G(V)$ is a division algebra. This means that every nonzero element is invertible. If k is algebraically closed, then $\mathrm{End}_G(V) \cong k$.

In the next few lectures, we will see that the very successful theory of characters is essentially a corollary of Schur's lemma. In particular, we will see that $\dim(\mathrm{Hom}_G(-, -))$ gives us a way to take an inner product of representations.

Example 2.5. As a last example, we will observe that if k is not algebraically closed, then $\mathrm{End}_G(V)$ need not be a 1-dimensional vector space over k, even if V is irreducible.

Consider the representation (V, ρ) given by

$$\begin{aligned} \rho : C_3 &\longrightarrow \mathrm{GL}_2(\mathbb{R}) \\ 1 &\longmapsto A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \end{aligned}$$

which is a representation over $\mathbb{R}$ since $A^3 = \mathrm{id}$. This representation is irreducible since the eigenvalues of A over $\mathbb{C}$ are nonreal. What does $\mathrm{End}_{C_3}(V)$ look like? It is the collection of matrices $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ such that $BA = AB$, which are exactly the matrices on the form

$$\begin{pmatrix} a & -b \\ b & a-b \end{pmatrix} = a \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \cdot \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

Thus,

$$\mathrm{End}_{C_3}(V) = \left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \right\rangle$$

has dimension 2. As a $\mathbb{R}$-algebra,

$$\mathrm{End}_{C_r}(V) \cong \mathbb{R}[e^{2\pi i/3}].$$

What is going on here? Recall that this representation is equivalent to a $\mathbb{R}[G]$-module V. When we tensor this with $\mathbb{C}$ we get a $\mathbb{C}[G]$-module $\mathbb{C} \otimes V$ which is decomposable as the direct sum of the representations $1 \mapsto e^{2\pi i/3}$ and $1 \mapsto e^{4\pi i/3}$. The lack of these two elements in the field $\mathbb{R}$ is what is making the situation more complicated.

Exercise 2.6. Let n be a positive integer.

(a) What are the irreducible representations of $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{C}$?
(b) Let p be a prime. What are the irreducible representations of $\mathbb{Z}/p\mathbb{Z}$ over $\mathbb{Q}$?
(c) What are the irreducible representations of $\mathbb{Z}/4\mathbb{Z}$ over $\mathbb{Q}$?

3. Lecture 3

In this lecture, we will explore applications of Schur's lemma. This forms the backbone of the theory of representations of finite groups over algebraically closed fields.

3.1. **Representations of abelian groups.**

Theorem 3.1. *Let G be an abelian group and k be an algebraically closed field. Then, every irreducible representation of G over k is 1-dimensional.*

Proof. Let $\rho : G \to \mathrm{GL}(V)$ be an irreducible representation. Since G is abelian we can check that for every $g \in G$, ρ_g is a morphism of representations $V \to V$. By Schur's lemma, $\rho_g = \lambda_g I$ for some scalar $\lambda_g \in k^\times$. Since any vector space is invariant under multiplication by scalars, it follows that if V is irreducible, then $\dim(V) = 1$. □

3.2. **General constructions.** Given two representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(V)$ of a group G over a field k, we already know one way to combine them, namely to take the direct sum. Unfortunately, this does not really give us any additional information about G and its representations. However, there are more fruitful things we can do.

For instance, we can form the *tensor product* $\rho \otimes \rho' : G \to \mathrm{GL}(V \otimes_k V')$ which is defined by $(\rho \otimes \rho')_g(v \otimes v') = \rho_g(v) \otimes \rho'_g(v')$. As a $k[G]$-module, this is of course simply $V \otimes_k W$.

We can consider a new representation $\psi : G \to \mathrm{GL}(\mathrm{Hom}(V, V'))$ given by

$$\psi_g(f) = \rho'_g \circ f \circ \rho_{g^{-1}}.$$

Lastly, if we have any group action of G on a finite set X, we can define an representation of G in $k[X]$ by:

$$\rho_h(\sum_g a_g \cdot g) = \sum_g a_{h^{-1}g} \cdot g.$$

The most interesting example is when $X = g$, in which case we get the *regular representation* $k[G]$ which is by far the most important representation as we shall see in the following section.

3.3. **Classification of irreducible representations.** Let $\rho : G \to \mathrm{GL}(V)$ be a representation. We leave it as a straightforward exercise to show the following important lemma.

Lemma 3.2. *We have the following isomorphism of vector spaces:*

$$\begin{aligned} \mathrm{Hom}_G(k[G], V) &\xrightarrow{\sim} V, \\ f &\longmapsto f(1_G). \end{aligned}$$

Notice that this is an isomorphism on the level of vector spaces and not between representations.

We now state our most important theorem.

Theorem 3.3. *Let G be a finite group and k a field of characteristic 0.*

(i) Every irreducible representation of G is isomorphically a subrepresentation of $k[G]$.
(ii) There are only finitely many irreducible representations of G up to isomorphism.
(iii) If k is algebraically closed and $P_1, \dots, P_m$ are all irreducible representations of G, then

$$\sum \dim(P_i)^2 = |G|.$$

Proof. By Maschke's theorem we can write

$$k[G] \cong \oplus_i P_i^{n_i}$$

for pairwise nonisomorphic irreducible representations P_i. By Lemma 3.2, for every representation V of G,

$$\begin{aligned} V &\cong \mathrm{Hom}_G(k[G], V) \\ &\cong \mathrm{Hom}_G(\oplus_i P_i^{n_i}, V) \\ &\cong \oplus_i n_i \mathrm{Hom}_G(P_i, V). \end{aligned}$$

By Schur's lemma, and $\dim(V) > 0$ we obtain that $P_i \cong V$ for some i. This proves (i) and (ii) follows immediately.

For (iii), note that $\dim(V) = n_i$ by Schur's lemma. Hence, (iii) follows from $\dim(k[G]) = |G|$. □

Example 3.4. We will now try to classify the irreducible representations for some symmetric groups over $\mathbb{C}$.

(a) Consider S_2. Since $2 = 1^2 + 1^2$ there are only two irreducible representations each of degree 1 and these are clearly the trivial representation and the sign representation.
(b) In general, let us find the 1-dimensional representations of S_n in two ways. One way is to notice that the abelianisation of G is $S_n/A_n \cong C_2$ and hence the only homomorphisms $G \to \mathbb{C}^\times$ are the trivial and sign map.

   Another way to see this is to consider the generators $s_k = (1 \quad k) \in S_n$. Finding a 1-dimensional representation requires solving the equations

$$\begin{aligned} s_i^2 &= 1 && \text{for } 1 \le i \le n \\ (s_i s_j)^3 &= 1 && \text{for } 1 \le i < j \le n \end{aligned}$$

over $\mathbb{C}^\times$, which is easily done and gives exactly the trivial representation and the sign representation.

(c) Consider S_3. Since $3! = 1^2+1^2+2^2$ we see that besides the 1-dimensional irreducible representations, we have a unique irreducible representation of dimension 2 which we have already seen is the symmetries of a regular triangle in the plane.

(d) For a general symmetric group S_n we clearly have a representation of S_n in $\mathbb{C}[X]$ where $X = \{1, \dots, n\}$. Furthermore, we see that this representation is isomorphic to a direct sum of the representations:

$$V = \langle (1, 1, \dots, 1) \rangle,$$
$$W = \{(x_1, \dots, x_n) : x_1 + \cdots + x_n = 0\}.$$

We claim that W is irreducible. Suppose not and that we have a nonzero proper subrepresentation $U \subseteq W$. This must contain a vector $(x_1, \dots, x_n)$ with $x_1 \neq x_2$. Hence, it also contains

$$(x_1, x_2, x_3, \dots, x_n) - (x_2, x_1, x_3, \dots, x_n) = (x_1 - x_2, x_2 - x_1, 0, \dots, 0).$$

It follows that

$$(1, -1, 0, \dots, 0), (1, 0, -1, 0, \dots, 0), \dots, (1, 0, \dots, 0, -1) \in U.$$

These vectors span V and hence $U = V$, a contradiction. Thus W is irreducible.

(e) Lastly, consider S_4. We have two irreducible 1-dimensional representations. By part (d), we have a 3-dimensional representation ρ and tensoring with the sign representation we get a new representation ρ'. These are non-isomorphic since the traces

$$\mathrm{Tr}(\rho_{(1\,2)}) = 1 \qquad \mathrm{Tr}(\rho'_{(1\,2)}) = -1$$

differ.

Since $24 = 1^2+1^2+3^2+3^2+2^2$ we have left to find one 2-dimensional irreducible representation. Notice that $V_4 \trianglelefteq S_4$ and hence we have surjection $S_4 \to S_4/V_4 \cong S_3$. Hence, the irreducible representation of S_3 gives us the unique irreducible representation of S_4 for free and our classification is completed.

Exercise 3.5. Prove that if we have a surjective group homomorphism $\varphi : G \to H$ and an irreducible representation $\rho : H \to \mathrm{GL}(V)$, then $\rho \circ \varphi : G \to \mathrm{GL}(V)$ is an irreducible representation.

Exercise 3.6. Let k be an algebraically closed field of characteristic 0 and G be a finite group. Prove that the number of 1-dimensional characters of G is equal to $|G/[G, G]|$.

Exercise 3.7. Let K/k be a field extension and let $\rho : G \to \mathrm{GL}(V)$ be a representation over k.

(a) Prove that

$$\dim(\mathrm{End}_{k[G]}(V)) = \dim(\mathrm{End}_{K[G]}(K \otimes V)).$$

(b) Deduce that if k and K are algebraically closed, then V is irreducible over k if and only if the representation $K \otimes_k V$ is irreducible over K.

4. Lecture 4

In this lecture, we will assume that k is an algebraically closed field of characteristic 0 and that G is a finite group with irreducible representations $P_1, \dots, P_m$. Our main goal is to prove that m is equal to the number of conjugacy classes of G.

4.1. **Module point of view.** Let us recall some terminology that we introduced in the first lecture. Recall that $k[G]$ is a k-algebra called the *group algebra* and that a representation of G in V over k is the same as a homomorphism of k-algebras:

$$k[G] \longrightarrow \mathrm{End}_k(V).$$

4.2. **Describing the group algebra.**

Theorem 4.1. *The map $\varphi : k[G] \longrightarrow \mathrm{End}_k(P_1) \times \cdots \times \mathrm{End}_k(P_m)$ is an isomorphism of k-algebras.*

Proof. Let $z \in k[G]$ be such that $\varphi(z) = 0$. Then, by Maschke's theorem, it follows that z acts trivially on any finite dimensional representation of G. In particular, it acts trivially on $k[G]$. This implies that $z \cdot e = 0$ and hence $z = 0$. This shows that φ is injective.

In the last lecture, we proved the formula $|G| = \sum \dim(P_i)^2$. This shows that

$$\dim(k[G]) = \dim(\mathrm{End}_k(P_1) \times \cdots \times \mathrm{End}_k(P_m))$$

and hence φ is an isomorphism of k-algebras, which was what we wanted to show. □

Example 4.2. Notice that the above property is special for *group algebras* and relies heavily on Maschke's theorem. In particular consider the k-algebra of *dual numbers* $k[\epsilon]/(\epsilon^2)$. Suppose that M is an irreducible module over this algebra and we have a map $\rho : k[\epsilon]/(\epsilon^2) \to \mathrm{End}_k(M)$. Note that $\ker \rho_\epsilon$ is stable and that ρ_ϵ is not injective. Thus, since M is irreducible, $\ker(\rho_\epsilon) = M$ and $M \cong k$. Thus, ϵ is a nonzero element of the group algebra acting trivially on all irreducible modules.

We can now deduce the main theorem of this lecture.

Theorem 4.3. *The number of isomorphism classes of irreducible representations of G over k is equal to the number of conjugacy classes of G.*

Proof. A quick computation shows that $\dim(Z(k[G]))$ is equal to the number of conjugacy classes in G. Additionally,

$$\dim(\mathrm{End}_k(P_1) \times \cdots \times \mathrm{End}_k(P_m)) = \sum_{i=1}^{m} 1 = m$$

since the center of a matrix algebra is 1-dimensional. The isomorphism in Theorem 4.1 finishes the proof. □

Exercise 4.4. Recall that we already showed using Schur's lemma that if G is abelian, then any irreducible representation has dimension 1. Prove the converse: if every irreducible representation of G has dimension 1, then G is abelian.

5. Lecture 5

In this lecture, we develop a computational tool for determining the irreducible representations of a finite group, namely *character theory.* Please note that there is not much new mathematical content in this lecture and that the theory of characters is essentially a useful corollary of the previous lecture. We will fix an algebraically closed field k of characteristic 0 and a finite group G. Furthermore, we will denote the irreducible representations of G over k by $P_1, \ldots, P_m$.

5.1. **Character theory.** Recall that we have an isomorphism of k-algebras:

$$k[G] \xrightarrow{\simeq} \prod_i \mathrm{End}(P_i).$$

In particular, this restricts to an isomorphism of vector spaces

$$A : Z(k[G]) \xrightarrow{\simeq} \prod_i k \cdot \mathrm{Id}.$$

Note that we have natural choices of bases on both the left and right side of the above isomorphism. Hence, we can try to understand concretely what the inverse of this map looks like. Additionally, recall that we can think of $Z(k[G])$ as functions on the conjugacy classes of G. Notice also that for any $\varphi \in \prod_i \mathrm{End}(P_i)$ can be interpreted as an endomorphism of $k[G]$, since

$$k[G] \cong P_1^{\dim(V_1)} \oplus \cdots \oplus P_m^{\dim(V_m)}.$$

Theorem 5.1. *The inverse of A is given by:*

$$B : \prod_i k \cdot \mathrm{Id} \longrightarrow Z(k[G]),$$

$$\varphi \longmapsto \frac{1}{|G|} \mathrm{Tr}(g^{-1} \circ \varphi : k[G] \to k[G]).$$

Proof. The proof follows from a computation. It is sufficient to verify that B is a left inverse of A. Indeed, for any $h \in G$:

$$\begin{aligned} B(A(h)) &= \frac{1}{|G|} \mathrm{Tr}(g^{-1} \circ h) \\ &= h \in k[G]. \end{aligned}$$

The latter equality follows from the fact that $g^{-1} \circ h$ permutes the natural basis elements of $k[G]$ and hence the trace above is $|G|$ when $g = h$ and 0 otherwise. □

Please note that there is some slight abuse of notation above. Let us now explore where each basis vector

$$E_i = (0, \ldots, \mathrm{id}, \ldots, 0)$$

is sent by B. For this we will need a new definition. If $\rho : G \to \mathrm{GL}(V)$ is a finite dimensional representation, then we define its *character* $\chi_\rho \in k[G]$ by $\chi_\rho(g) = \mathrm{Tr}(\rho_g)$ which is well-defined since the trace is invariant under conjugation.

Now, we can note that:

$$\begin{aligned} B(E_i) &= \frac{1}{|G|} \mathrm{Tr}(g^{-1} \circ E_i) \\ &= \frac{\dim(P_i)}{|G|} \mathrm{Tr}(g^{-1} : P_i \to P_i) \\ &= \frac{\dim(P_i)}{|G|} \chi_{P_i}(g^{-1}). \end{aligned}$$

We can define a symmetric inner product on $k[G]$ by

$$(f, h) = \frac{1}{|G|} \sum_g f(g) \cdot h(g^{-1})$$

for $f, h \in G$.

Theorem 5.2. *We have the the following results:*

(i) $(\chi_{P_i}, \chi_P) = \delta_{ij}$,
(ii) $\rho : G \to \mathrm{GL}(V)$ *is irreducible if and only if* $(\chi_\rho, \chi_\rho) = 1$,
(iii) $\rho : G \to \mathrm{GL}(V)$ *and* $\rho' : G \to \mathrm{GL}(W)$ *are isomorphic if and only if* $\chi_\rho = \chi_{\rho'}$.

Proof. Obviously $E_iE_j = \delta_{ij}E_i$ in $\prod_i \mathrm{End}(P_k)$. In $k[G]$, this translates To

$$\left(\sum \chi_{\rho_i}(g^{-1})g\right)\left(\sum \chi_{\rho_j}(g^{-1})g\right) = \delta_{ij}\frac{|G|}{\dim(P_i)}\sum \chi_{\rho_i}(g^{-1})g.$$

Equating the coefficients of $g = 1$, we obtain that

$$\sum \chi_{\rho_i}(g)\chi_{\rho_j}(g^{-1})g = \delta_{ij}|G| \iff (\chi_{\rho_i}, \chi_{\rho_j}) = \delta_{ij},$$

which proves (i). Notice that if $\rho : G \to \mathrm{GL}(V)$, then there exist nonnegative integers n_i such that $V \cong \oplus_i P_i^{n_i}$. Then,

$$(\chi_\rho, \chi_\rho) = \sum_i n_i^2$$

is equal to 1 if and only if one n_i is 1 and the rest are 0. This proves (ii). Moreover,

$$(\chi_\rho, \chi_{P_i}) = n_i$$

which proves (iii). □

The next theorem shows that we can obtain a canonical projection of a representation $V \cong \oplus P_i^{n_i}$ onto $P_i^{n_i}$. This is referred to as a canonical decomposition into *isotypical components*. It is a straightforward exercise and will be left to the reader.

Theorem 5.3. *Consider*

$$\pi_i := \frac{\dim(P_i)}{|G|}\sum_g \chi_{\rho_i}(g^{-1})g \in k[G].$$

Then π *acts like the identity on* P_j *when* $i = j$ *and like* 0 *when* $i \neq j$. *Furthermore, for any representation* V,

$$\rho_{\pi_i} : V \to V$$

is a G*-homomorphism and a projection onto the direct sum of all the copies of* P_i *in* V.

Let us consider one of the simplest examples of the above theorem.

Example 5.4. Let $G = \mathbb{Z}/n\mathbb{Z}$. Let $\omega = e^{2\pi i/n}$ and consider the irreducible characters of G given by

$$\begin{aligned} \rho_j : G &\longrightarrow \mathbb{C}^\times \\ 1 &\longmapsto \omega^j. \end{aligned}$$

Then, given any representation $V = \oplus P_i^n$, we have the projections:

$$\rho_{\pi_i}(v) = \frac{1}{n}\sum_{\bar{m}\in\mathbb{Z}/n\mathbb{Z}} \omega^{-m}\cdot\rho_{\bar{m}}(v).$$

For example, when $n = 2$, we have the two projections

$$\rho_{\pi_0} = \frac{1}{2}(\rho_0 + \rho_1) \quad \text{and} \quad \frac{1}{2}(\rho_0 - \rho_1).$$

As a special case, if W is any vector space, then we have a representation of $\mathbb{Z}/2\mathbb{Z}$ on $W \otimes W$ given by

$$\rho_1(w_1 \otimes w_2) = w_2 \otimes w_1.$$

Using this, we get a decomposition

$$W \otimes W = S^2W \oplus \Lambda^2 W$$

where S^2W denotes the symmetric tensors and $\Lambda^2 W$ the alternating tensors.

5.2. **Character tables.** Given a group G, we can construct its character table. Two examples are given below.

Example 5.5. Let $G = S_3$. We have already constructed all the irreducible representations in earlier lectures and we therefore see that the we have the character table below.

	1	(1 2)	(1 2 3)
trivial	1	1	1
sign	1	-1	1
symmetries of a triangle	2	0	-1

We will soon proceed to a more complicated example, but we will first need a lemma.

Lemma 5.6. *Suppose that $H \leq G$ is an abelian subgroup. Then any irreducible representation of G has order $\leq \frac{|G|}{|H|}$.*

Proof. Let V be an irreducible representation of G. As a representation of H, it has a 1-dimensional subrepresentation W. Then, if $G = \cup_i g_i H$, we see that $\sum \rho_{g_i}(W) \subseteq V$ is a subrepresentation of G. Since V is irreducible we have that $V = \sum \rho_{g_i}(W)$ and hence $\dim(V) \leq |G|/|H|$ as claimed. $\square$

Example 5.7. Let $G = D_4$ and recall that $|D_4| = 8$. Since $C_4 \leq D_4$ all irreducible representations of D_4 are of dimension ≤ 2. To find the 1-dimensional representations of D_4, we simply have to solve the system of equations

$$r^4 = s^2 = 1,$$
$$srs = r^{-1}.$$

which gives solutions $r, s \in \{\pm 1\}$. Thus, all 1-dimensional representations are real and since the 2-dimensional representation of D_4 as symmetries of a square is irreducible over $\mathbb{R}$, it is clear that it is also irreducible over $\mathbb{C}$ for this reason. Hence, we obtain the character table below.

e	$\{r, r^3\}$	$\{r^2\}$	$\{s, sr^2\}$	$\{sr, sr^3\}$
1	1	1	1	1
1	-1	1	1	-1
1	-1	1	-1	1
1	1	1	-1	-1
2	0	-2	0	0

5.3. **Constructing new representations.** One might be misled by character theory to think that the subject of the representation theory of finite groups is trivial. This is not the case. The hard part is to generate enough irreducible representations. One fruitful way we have seen to construct new representations is by taking hte tensor product and we shall see that the character of the tensor product is easy to compute.

Proposition 5.8. *If $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$ are representations, then the character of $\rho \otimes \rho' : G \to GL(V)$ is given By*

$$\chi_{\rho\otimes\rho'}(g) = \chi_\rho(g) \cdot \chi_{\rho'}(g).$$

The proof follows from the below lemma from linear algebra.

Lemma 5.9. *If V and W are finite-dimensional vector spaces and $A \in \mathrm{End}(V)$ and $B \in \mathrm{End}(W)$, then $\mathrm{Tr}(A \otimes B) = \mathrm{Tr}(A) \otimes \mathrm{Tr}(B)$.*

Proof. To prove this, note that both the left and right hand side are linear in A and in B. Thus, it is sufficient to prove this claim when $A = E_{ij}$ and $B = E'_{i'j'}$ for some elementary matrices. This is an easy computation. □

6. Lecture 6

6.1. **Constructing new representations.** Last lectire. we discussed how to obtain new representations from old by taking the tensor product. Today, we will give a couple of further examples. Given a representation $\rho : G \to \mathrm{GL}(V)$, we define the *dual representation* $\rho^* : G \to \mathrm{GL}(V^*)$ by

$$(\rho^*_g(f))(v) := f(\rho_g^{-1}(v))$$

for each $f \in V^*$, $v \in V$ and $g \in G$. If we choose a basis for V and its dual basis for V^*, we obtain a map

$$\begin{aligned} \mathrm{GL}_n(k) &\longrightarrow \mathrm{GL}_n(k) \\ A &\longmapsto (A^t)^{-1} \end{aligned}$$

such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\rho"] \arrow[dr, "\rho^*"'] & \mathrm{GL}_n(k) \arrow[d] \\ & \mathrm{GL}_n(k) \end{tikzcd}$$

Hence, we see that the character of ρ^* is

$$\chi_{\rho^*}(g) = \chi_\rho(g^{-1})$$

for $g \in G$.

Moving forward we let $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$ be representations. Notice that $\mathrm{Hom}_k(V, W) \cong V^* \otimes W$ when V is finite-dimensional. Using that $V^* \otimes W$ is a representation of G, we get that $\mathrm{Hom}_k(V, W)$ is a representation of G and that it is given By

$$\rho'' : G \to \mathrm{Hom}_k(V, W)$$
$$\rho''_g(f)(v) = \rho'_g(f(\rho_g^{-1}(v))).$$

Note that we can define $V^G = \{v \in V : \rho_g(v) = v \text{ for all } g \in G\} \subseteq V$ and that G acts trivially on this set.

Exercise 6.1. Show that $\mathrm{Hom}_k(V,W)^G = \mathrm{Hom}_G(V,W)$.

We can now show a theorem that generalises the orthogonality relations for irreducible representations over an algebraically closed field of characteristic 0.

Theorem 6.2. *Let k be a field such that $\mathrm{char}(k) \not| |G|$ and consider two finite-dimensional representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$. Then*

$$(\chi_\rho, \chi_{\rho'}) = \dim_k \mathrm{Hom}_G(W,V).$$

Proof. We first remark that that for any representation $\rho'' : G \to \mathrm{GL}(Q)$, the map

$$\pi = \frac{1}{|G|} \sum_{g \in G} \rho''_g$$

is a projection onto Q^G. Thus,

$$\begin{aligned}
(\chi_\rho, \chi_{\rho'}) &= \frac{1}{|G|} \sum_{g \in G} \chi_\rho(g) \cdot \chi_{\rho'}(g^{-1}) \\
&= \frac{1}{|G|} \sum_{g} \chi_{W^* \otimes V}(g) \\
&= \mathrm{Tr}\Big(\frac{1}{|G|} \sum_{g \in G} \rho_{W^* \otimes V}(g)\Big) \\
&= \dim((W^* \otimes V)^G) \\
&= \dim(\mathrm{Hom}_G(W,V))
\end{aligned}$$

using the previous remark. $\square$

Exercise 6.3. Verify some of the linear algebra facts we used in the proof of Theorem 6.2.

(a) Prove that π in the theorem is a projection.
(b) Prove that for any linear projection $\pi : V \to V$, we have that $\mathrm{Tr}(\pi) = \dim(\pi(V))$.

Exercise 6.4. Prove that if $k = \mathbb{C}$, then $\chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$.

6.2. **Representations of products.** In this subsection, we consider a field $\mathbb{Q} \subseteq k = \bar{k}$. We shall show that to understand the representations of a product of groups, it is sufficient to understand the representations of the components. Note first that if G_1 acts on V_1 and G_2 acts on V_2, then $G_1 \times G_2$ acts naturally on $V_1 \otimes_k V_2$.

Proposition 6.5. *If $P_1, \ldots, P_d$ are the irreducible representations of G_1 and $Q_1, \ldots, Q_m$ are the irreducible representations of G_2, then all irreducible representations of $G_1 \times G_2$ are given by $P_i \otimes Q_j$.*

Proof. Notice firstly that

$$\begin{aligned}
(\chi_{P_i \otimes Q_j}, \chi_{P_{i'} \otimes Q_{j'}}) &= \frac{1}{|G_1| \cdot |G_2|} \sum_{g_1 \in G_1,\, g_2 \in G_2} \chi_{P_i}(g_1) \cdot \chi_{Q_j}(g_2) \cdot \chi_{P_{i'}}(g_1^{-1}) \cdot \chi_{Q_{j'}}(g_2^{-1}) \\
&= \left(\frac{1}{|G_1|} \sum_{g \in G_1} \chi_{P_i}(g_1) \cdot \chi_{P_{i'}}(g_1^{-1}) \right) \cdot \left(\frac{1}{|G_2|} \sum_{g \in G_2} \chi_{Q_j}(g_2) \cdot \chi_{Q_{j'}}(g_2^{-1}) \right) \\
&= (\chi_{P_i}, \chi_{P_{i'}}) \cdot (\chi_{Q_j}, \chi_{Q_{j'}}).
\end{aligned}$$

This shows that

$$(\chi_{P_i \otimes Q_j}, \chi_{P_{i'} \otimes Q_{j'}}) = \begin{cases} 1 & \text{if } P_i \cong P_{i'} \text{ and } Q_j \cong Q_{j'} \\ 0 & \text{else.} \end{cases}$$

In other words, each $P_i \otimes Q_j$ is irreducible and these representations are pairwise nonisomorphic. Since the number of conjugacy classes of $G_1 \times G_2$ is the product of the number of conjugacy classes of G_1 and G_2 respectively, we have found them all. Thus, the proof is completed. $\square$